Technische Universität München

Business Forecasting Final Report

Sales Forecasting Framework for Retail Operations

Yu-Ching Lin, Xueying Peng

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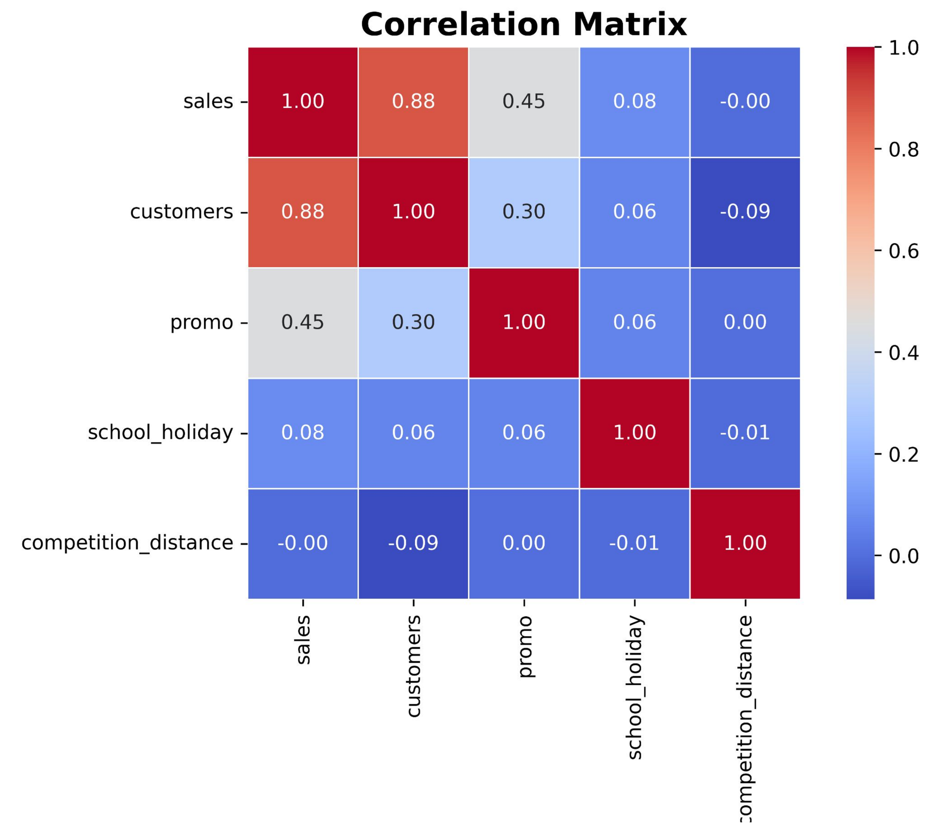
Target: Develop an accurate and maintainable sales forecasting framework for weekly sales prediction across a multi store retail chain.

EDA and Feature Engineering

Data Cleaning :Chronologically aligned all store records and standardized forecasting variables for consistent multi store time series modeling. Also checking and dealing with duplicates and missing values.

Correlation Analysis:Correlation analysis highlighted customers (0.88) and promotions (0.45) as the strongest variables related sales , guiding the 2 Random Forest Model decision.

Feature Engineering: With reference of Correlation, Constructing lag, rolling, calendar, promotion, and store-level features to capture temporal patterns and operational effects.



Benchmark Result

Benchmark Selection:Compared six forecasting methods (Mean, Naive, Drift, Seasonal Naive, AutoARIMA, and ETS) primarily using MAPE.

Model Evaluation: ETS achieved the **best performance in 449 of 676 stores (66.4%)**, consistently outperforming the other benchmark models.

Final Model:ETS was selected as the final benchmark for robustness across store types, and operational simplicity for large scale deployment.

Model	Stores Won	Win Rate (%)
ETS	**449**	**66.42**
Mean	135	19.97
AutoARIMA	88	13.02
Seasonal Naïve	4	0.59

Random Forest

Two-stage Forecasting Framework

Two stage Forecasting Process

Stage 1: Forecast customer numbers with Global RF.

Stage 2: Forecast sales using predicted customers and operational features.

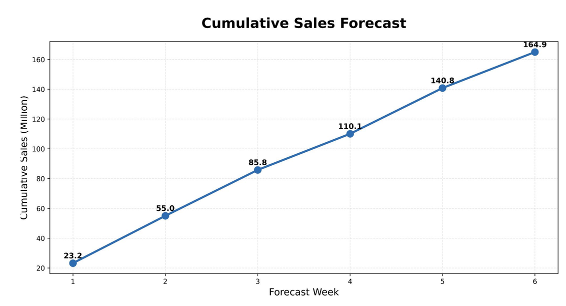
Forecasting Task	Model	Trees	Max Features	Min Leaf	MAPE (%)	Selected
Customer	RF200_SQRT	200	sqrt	1	**8.63**	✓
Customer	RF300_SQRT_Leaf5	300	sqrt	5	8.77	
Sales	RF200_SQRT	200	sqrt	1	**7.82**	✓
Sales	RF300_SQRT_Leaf5	300	sqrt	5	8.04	

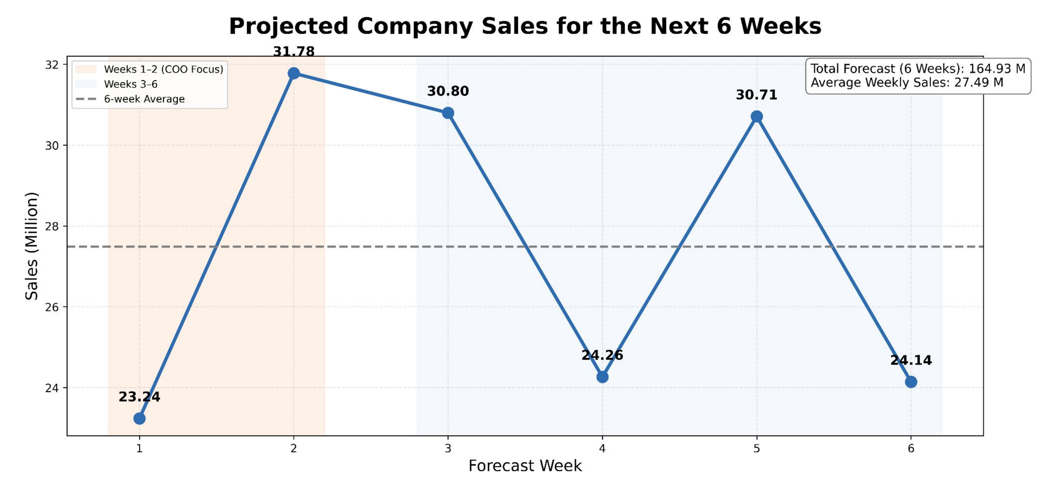
Model	MAPE (%)	Improvement
ETS	**17.07**	—
Global RF (without customer)	10.12	↓40.7%
Final Global RF (with customer)	**7.82**	↓54.2%

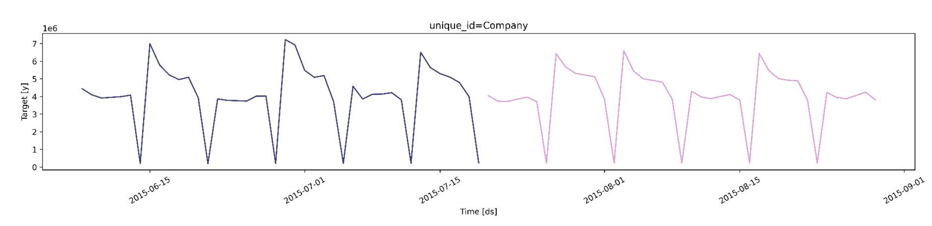
Result: 54.2% lower MAPE compare to the ETS baseline.

Overall Forecast Result

TOTAL SALES FORECAST
NEXT 6 WEEKS
164.9 M
(173.8 in ETS)
Company-wide Forecast





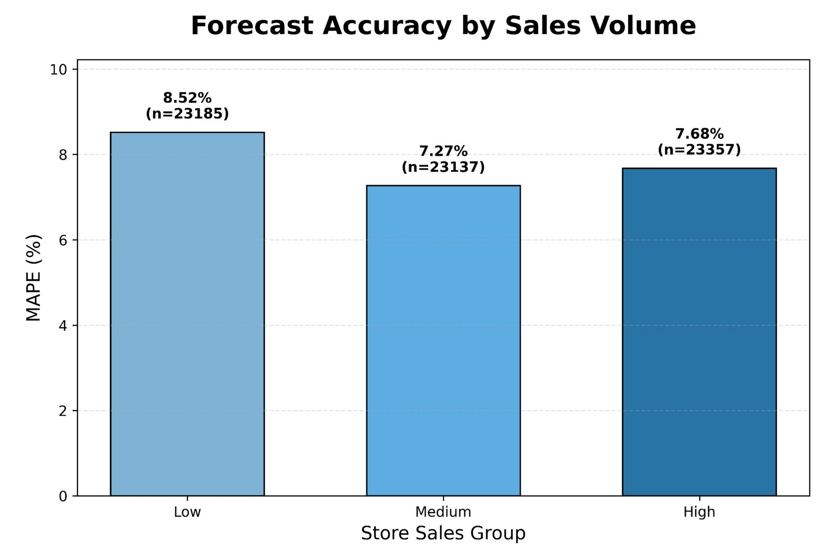


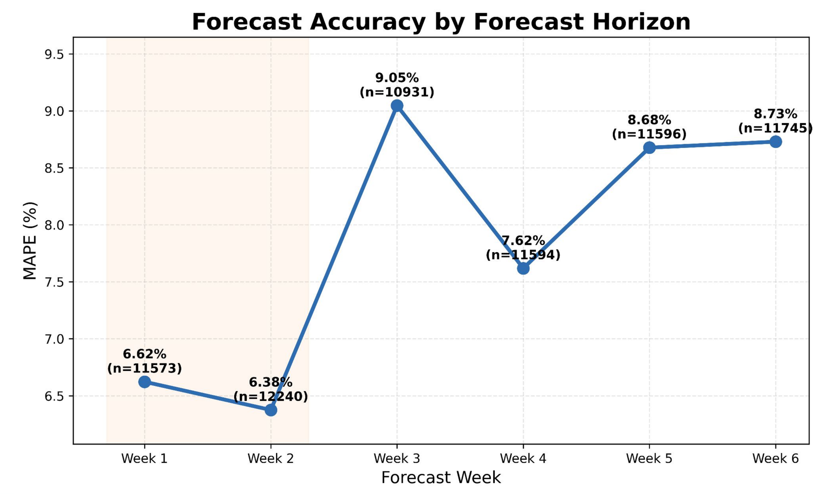
Stable revenue: Weekly income shows little fluctuation, resulting in a smooth cumulative growth curve.

Cycle period: There are similar cycles appearing through whole horizon time.

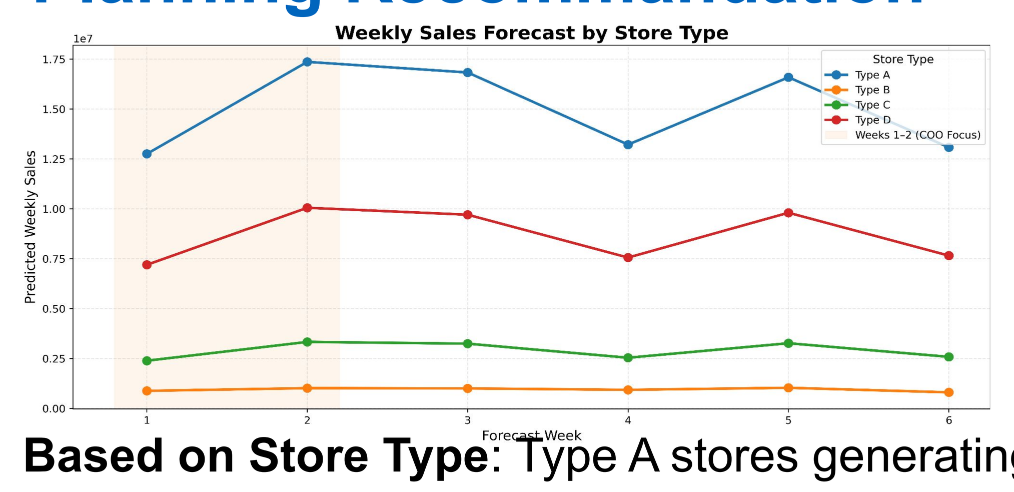
=>We can predict **relatively stable cash flow with cyclical pattern** in future.

Forecast Accuracy Analysis





Planning Recommendation

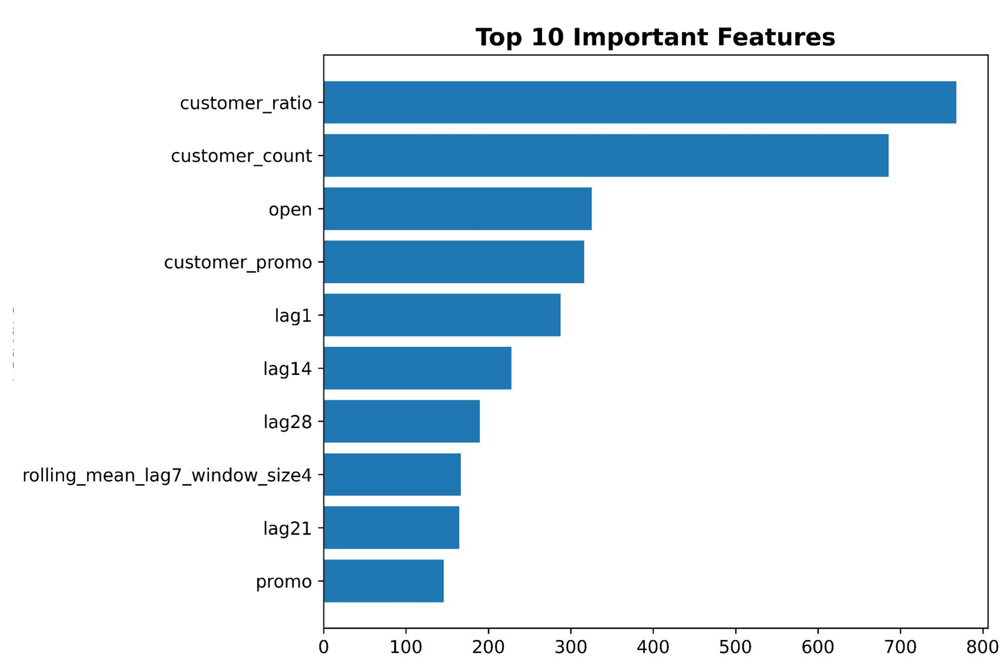


Based on Store Type: Type A stores generating the highest forecast sales, B stores contribute substantially less.

Sales are concentrated in a few stores:**Top 20** stores generate **6.7%** of forecast sales.

=>We could prioritise inventory and delivery resources , and make replenishment strategies

Feature Importance



Customer-related features have the highest predictive power.Accurate forecasts rely on data quality of important features.

Maintenance

Weekly Monitoring: Retrain the Global Random Forest weekly with new data, checking data quality and tracking metrics.

Limitations and Future Improvements

Have longer historical customer data and add detailed market information, like store density, urban–rural characteristics to better capture demands.Also Develop store segmentation or standard model rules .